CHIWOO ROH

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EDUCATION

Sep. 2023 - Aug. 2025 **Ajou University** Suwon, Korea
Department of Data/Network/AI Convergence (Transportation Engineering Program)
Master of Science in Transportation Engineering
Thesis: LLM-Based Contextual Understanding of Real-World Driving Behavior and Realistic Reproduction in Traffic Simulation
Advisor: Dr. Jaehyun So

Mar. 2018 - Aug. 2023 **Ajou University** Suwon, Korea
Department of Transportation Systems Engineering
Bachelor of Transportation Systems Engineering

RESEARCH INTERESTS

Traffic Operations, Traffic Safety, Driving Behavior Modeling, Human Mobility Prediction, Generative AI for Human Mobility and Driving Behavior, Artificial Intelligence in Transportation

TEACHING EXPERIENCES

Lecturer

1. Application of Autonomous Driving Simulation Technology (Practical Training - SUMO), 2024 ITS & Smart Mobility Summer School Aug. 2024

Teaching Assistant

1. Mobility Seminar I [O1391], Ajou University Fall 2024
2. Mobility Data Analytics [E091], Ajou University Spring 2024
3. Highway Capacity Analysis [E092], Ajou University Spring 2024
4. Smart Mobility Service [E046], Ajou University Fall 2023
5. Traffic Study and Data Analytics [E051], Ajou University Fall 2023
6. Unstructured Data Analysis Theory and Practice, Hyundai Motor Group 2023 - 2024

RESEARCH EXPERIENCES

Researcher at *Korea Transport Institute* Jul. 2025 - Present
- Human Mobility Prediction Using LLMs with Individual Travel Surveys
- Development of AI-Based Individual-Level Mobility Data Synthesizing for Privacy Protection
- Development for Measuring Congestion of Gathering Occasion using Mobile Phone Location Data

Research Assistant at *Movelab, Ajou University* Jan. 2023 - Jun. 2025
- Evaluation of Autonomous Vehicle Legal Compliance Using Traffic Simulation
- Developing Hazardous Driving Indicators and Determination Algorithm Based on Transport Big Data

Visiting Student at *Texas A&M University* (Korea-US NSF IRES : AIxMobility) Jan. 2024 - Feb. 2024
- Nighttime Road Image Correction System Using Generative Adversarial Networks

PUBLICATIONS

Journals

1. **Roh, C.**, & Won, M. (2026). Synthesizing Human Mobility Patterns Using Large Language Models - A Sejong City Case Study. Journal of Korean Society of Transportation, 45(2).
2. You, J., **Roh, C.**, Kim, J., Lee, E., & So, J. (2026). A Methodological Framework for Driving Behavior Analysis Based on Real-World Trajectory Data. International Journal of Highway Engineering, 28(1).
3. Cho, S., **Roh, C.**, You, S., & So, J. (2024). Comparative analysis of the 3D object detection performance of LiDAR sensors for traffic monitoring. International Journal of Highway Engineering, 26(5), 143–150.
4. Kim, H., Kim, J., Lee, S., & **Roh, C.** (2024). Analysis of automation level characteristics of autonomous driving simulation research. Journal of the Korean Society of Intelligent Transport Systems, 23(6), 366–381.

Conference Proceedings

1. **Roh, C.**, & Won, M. (2026). Development of a Method for Synthesize of Individual-Level Travel Chains Using Large Language Models. In 2026 Korean Society of ITS Spring Conference, Jeju, South Korea. The Korean Society of Intelligent Transport Systems.
2. **Roh, C.**, Cho, S., Won, M., & So, J. (2026). Contextual modeling of real-world driving behavior with large language models and realistic reproduction in traffic simulation. In Transportation Research Board (TRB) 105th Annual Meeting, Washington, DC, United States. Transportation Research Board(TRB)
3. **Roh, C.**, Cho, S., Park, H., Park, J., & So, J. (2026). A hybrid framework for traffic crash risk analysis using crash data and microscopic simulations. In Transportation Research Board (TRB) 105th Annual Meeting, Washington, DC, United States. Transportation Research Board(TRB)
4. Cho, S., Ryoo, S., **Roh, C.**, Lee, S., Betz, J., Margreiter, M., & So, J. (2026). LiDAR on the Move: Extracting Lane Level Traffic Indicators from Autonomous Vehicles. In Transportation Research Board (TRB) 105th Annual Meeting, Washington, DC, United States. Transportation Research Board(TRB)
5. You, S., Park, H., **Roh, C.**, Cho, S., & So, J. (2026). Multi-Modal Trajectory Prediction of Child Pedestrians Using a Unified Transformer Model. In Transportation Research Board (TRB) 105th Annual Meeting, Washington, DC, United States. Transportation Research Board(TRB)
6. **Roh, C.**, Cho, S., Park, H., & So, J. (2025). A hybrid traffic crash risk analysis modeling of crash data-driven analysis and microscopic simulations. In 2025 IEEE 28th International Conference on Intelligent Transportation Systems (ITSC), Gold Coast, Australia. IEEE.
7. Cho, S., Ryoo, S., **Roh, C.**, Lee, S., & So, J. (2025). Monitoring and measurement estimation using LiDAR sensors on autonomous vehicles. In 2025 IEEE 28th International Conference on Intelligent Transportation Systems (ITSC), Gold Coast, Australia. IEEE.
8. **Roh, C.**, Lee, J., & Won, M. (2025). Synthesizing human mobility patterns using retrieval-augmented LLMs: A Sejong City case study. In 2025 Korean Society of Transportation Fall Conference, Jeju, South Korea. The Korean Society of Transportation
9. **Roh, C.**, & So, J. (2025). Clustering-based aggressive driving analysis and LLM-driven trajectory generation. In 2025 Suwon ITS Asia-Pacific Forum, Suwon, South Korea ***(Academic Award for the Best Paper).***
10. **Roh, C.**, Cho, S., & So, J. (2025). LLM-driven trajectory synthesis and driving behavior analysis. In 2025 Korean Society of ITS Spring Conference, Jeju, South Korea. The Korean Society of Intelligent Transport Systems.
11. You, S., Park, H., **Roh, C.**, Cho, S., & So, J. (2025). Analysis of traffic flow changes due to abnormal vehicle stops in incident situations: A microscopic approach to spatial impact zones. In 2025 Korean Society of ITS Spring Conference, Jeju, South Korea. The Korean Society of Intelligent Transport Systems.
12. **Roh, C.**, Lee, E., & So, J. (2025). Personality on the road: Driving behavior analysis and trajectory generation with LLMs. In 2025 Korean Society of Transportation Spring Conference, Seoul, South Korea. The Korean Society of Transportation. ***(Academic Award for the Best Paper).***

13. Cho, S., **Roh, C.**, You, S., & So, J. (2025). A proof-of-concept study for LiDAR-based traffic measure estimation: Traffic density estimation and vehicle trajectory extraction. In 2025 Korean Society of Transportation Spring Conference, Seoul, South Korea. The Korean Society of Transportation.
14. Lee, Y., Kim, J., **Roh, C.**, & So, J. (2024). Cooperation and competition among vehicles in autonomous driving merge scenario: A proof-of-concept study. In 2024 Korean Society of ITS Fall Conference, Gangneung, South Korea. The Korean Society of Intelligent Transport Systems.
15. You, S., Park, H., **Roh, C.**, Cho, S., & So, J. (2024). Enhancing traffic safety through predicting child pedestrian trajectories: A study using transformer-based deep learning approach. In 2024 Korean Society of Transportation Fall Conference, Gangneung, South Korea. The Korean Society of Transportation.
16. Seo, Y., Park, J., Oh, G., Kim, H., **Roh, C.**, Hu, J., & So, J. (2024). Extracting traffic crash information from imbalanced-unstructured traffic crash description texts using a text classification modeling technique. In Transportation Research Board (TRB) 103rd Annual Meeting, Washington, DC, United States. Transportation Research Board(TRB)
17. **Roh, C.**, Lee, E., & So, J. (2023). A study on evaluation of autonomous vehicle legal compliance using traffic simulation. In 2023 Korean Society of ITS Fall Conference, Jeju, South Korea. The Korean Society of Intelligent Transport Systems.
18. **Roh, C.**, Lee, E., & So, J. (2023). A study on evaluation indicators for compliance with autonomous vehicle legislation: Focusing on the speed limit adherence indicator. In 2023 Joint Conference of the Korean Society of Transportation & Korea Planning Association, Busan, South Korea. The Korean Society of Transportation.
19. **Roh, C.**, Lee, E., & So, J. (2023). Development of self-driving vehicle evaluation interface using traffic simulation. In 2023 Korean Society of ITS Spring Conference, Yeosu, South Korea. The Korean Society of Intelligent Transport Systems.

Papers in Preparation

1. **Roh, C.**, & Won, M. (2026). Development of a Method for Synthesize of Individual-Level Travel Chains Using Large Language Models. Journal of Korean Society of Intelligent Transport Systems. *(Under Review)*
2. Cho, S., Ryoo, S., **Roh, C.**, Lee, S., Betz, J., Margreiter, M., & So, J. (2026). AVs as Mobile Traffic Sensors: Lane-Level Traffic State and Risk Indicator Extraction Using Vehicle-Mounted LiDAR. IEEE Transactions on Intelligent Transportation Systems. *(Under Review)*
3. You, S., Park, H., Cho, S., **Roh, C.**, & So, J. (2026). Predicting Child Pedestrian Behavior with Transformer-based Multi-agent Trajectory Prediction. IEEE Access. *(Under Review)*
4. Park, H., **Roh, C.**, Hu, J., Jia, W., Lee, J., Park, J., & So, J. (2025). Predicting traffic crash severity: A BERT-based modeling approach for integrating structured and unstructured data. IEEE Transactions on Intelligent Transportation Systems. *(Under Review)*

PATENTS

Video Data-Based Method and Device for Estimating Exhaust Vehicle Emissions

- Applicant: Ajou University
- Inventor: Jaehyun So, Joonho Park, **Chiwoo Roh**, Sungwon Kim, Woosung Na, Jaemo Ha, Minseo Kim
- Country: Republic of Korea
- Status: Applicant (10-2025-0197728)

Method and Apparatus for Regulatory Adaption of Autonomous Driving Logic

- Applicant: Ajou University
- Inventor: Jaehyun So, **Chiwoo Roh**, Eugene Lee
- Country: Republic of Korea
- Status: Applicant (10-2024-0187517)

Autonomous Vehicle Logic Generation Device and Method for Traffic Simulation

- Applicant: Ajou University
- Inventor: Jaehyun So, **Chiwoo Roh**, Eugene Lee
- Country: Republic of Korea
- Status: Applicant (10-2023-0163241)

AWARDS

- Academic Award for the Best Paper, Intelligent Transportation System Asia-Pacific Forum Secretariat, May. 2025
- Academic Award for the Best Paper, The Korean Society of Transportation, Feb. 2025
- Korean Transportation Association in America (KOTAA) Travel Grant Award, Korean Transportation Association in America, Jan. 2024

PROJECTS

Development of AI-Based Individual-Level Mobility Data Synthesizing for Privacy Protection

- Funder : Ministry of Land Infrastructure and Transport
- Program : Support Program for Emerging Researchers in Science and Technology in Land Infrastructure, and Transport
- Project Duration : Apr. 2026 - Dec. 2028 (Participated: Apr. 2026 - Present)
- Funding : 670 million KRW ($453,421 (USD))
- Role : Researcher (at Korea Transport Institute)

Technical Development for Measuring Congestion of Gathering Occasion using Mobile Phone Location Data

- Funder : Ministry of Science and ICT / Korean National Police Agency
- Program : Project for the Commercialization of Public Research Outcomes in Science-Based Policing
- Project Duration : Jul. 2023 - Dec. 2026 (Participated: Apr. 2026 - Present)
- Funding : 1,200 million KRW ($812,100 (USD))
- Role : Researcher (at Korea Transport Institute)

Establishment of Real-time Open Mobility Service Platform and Industrial Ecosystem

- Funder : Ministry of Trade and Industry
- Program : Regional Innovation Cluster Develop Project
- Project Duration : Jun. 2023 - Dec. 2027 (Participated: Jul. 2025 - Present)
- Funding : 6,730 million KRW ($4.8 million (USD))
- Role : Researcher (at Korea Transport Institute)

A Study on De-identification Technologies and Legal System Improvements for Privacy Protection of Object-level Mobility Data

- Funder : Korea Transport Institute
- Program : Institutional Research Project
- Project Duration : Apr. 2025 - Mar. 2026 (Participated: Jul. 2025 – Mar. 2026)
- Funding : 75 million KRW ($53,380 (USD))
- Role : Researcher (at Korea Transport Institute)

Graduate Program for Convergence Technology of Data/Network/AI and Road Transport

- Funder : Ministry of Land Infrastructure and Transport
- Program : Graduate Program for Convergence Technology of Data/Network/AI
- Project Duration : May. 2022 - Dec. 2027 (Participated: Jun. 2023 - Jun. 2025)
- Funding : 6,140 million KRW ($4.4 million (USD))
- Role : Research Assistant (at Ajou University)

Developing Hazardous Driving Indicators and Determination Algorithm Based on Big Data Technology

- Funder : Hyundai Motor Company
- Program : Hyundai Motor Company Industry-University Cooperation Program
- Project Duration : Jun. 2024 - Jun. 2025 (Participated: Jun. 2024 - Jun. 2025)
- Funding : 100 million KRW ($71,110 (USD))
- Role : Research Assistant (at Ajou University)

Development of Technology for Validating the Autonomous Driving Services in Perspective of Laws and Regulations

- Funder: Ministry of Science and ICT
- Program : Korea Autonomous driving Development Innovation Project
- Project Duration : Apr. 2021 - Dec. 2024 (Participated: Jan. 2023 - Dec. 2024)
- Funding : 4,100 million KRW ($2.9 million (USD))
- Role : Research Assistant (at Ajou University)

Research to Improve Mechanical Parking Lot Legislation

- Funder: Korea Transportation Safety Authority
- Project Duration : Mar. 2023 - Dec. 2023 (Participated: Mar. 2023 - Dec. 2023)
- Funding : 39 million KRW ($27,760 (USD))
- Role : Research Assistant (at Ajou University)

SKILLS & OTHERS

- Programming: Python (Pandas, PyTorch, TensorFlow, etc.), C/C#, MATLAB
- Tools: SUMO, VISSIM, QGIS, MS Office
- Languages: Korean (Native), English (Advanced)